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ABSTRACT

Objective: To evaluate the association between sociodemographic and economic factors and levels of Knowledge, Attitude, and Practice (KAP) regarding the use of regional foods by families with children under five years of age in rural Ceará. **Method:** This is a cross sectional study conducted with 70 mothers residing in the municipalities of Acarape/CE and Redenção/CE between January and March 2023. A sociodemographic questionnaire and the KAP survey were used to assess the domains associated with the use of regional foods. **Results:** Most participants had inadequate knowledge (82.9%), while 81.4% had an adequate attitude, and 75.7% reported adequate eating practices. There was a statistically significant association between the maternal occupation "housewife" and adequate eating practices ($p < 0.05$). Multivariate analysis showed that participants informed about nutrition were 26.7% more likely to have adequate knowledge ($OR = 0.733$, $p < 0.05$), and mothers with an adequate attitude were 5.4 times more likely to have adequate eating practices ($OR = 5.483$, $p < 0.05$). **Conclusion:** Sociodemographic and economic factors, with an emphasis on occupation and access to information about child nutrition, directly influence KAP in terms of the use of regional foods. These findings reinforce the need for educational strategies and public policies to promote healthy eating in early childhood.

Keywords: Knowledge, Attitudes, and Practices in Health; Eating Habits; Socioeconomic Factors; Child Health; Nursing

INTRODUCTION

Early childhood is a crucial period for the formation of eating habits that tend to last throughout life. Reduced consumption of fresh or minimally processed foods, such as regional foods, can compromise the supply of essential nutrients, promotes child food insecurity, and is associated with consequences that can extend into adulthood, such as an increased risk of iron deficiency anemia, vitamin A deficiency, and deficits in child growth and development¹.

foods are those that are widely available in different regions of Brazil, are , low cost, and easily accessible, making them important allies for food and nutritional security²⁻⁴. Despite these benefits, national surveys have shown that regional foods are becoming less and less common in Brazilian households. In the 2017-2018 period, typically regional foods accounted for only a quarter of the food groups consumed and represented 3.12% of total calories. It should be noted, however, that the Northeast region has the highest rates of consumption of regional foods⁵.

Dietary patterns are influenced by sociodemographic context. Rural residents, for example, consume large amounts of minimally processed foods, such as beans, fruits, and vegetables; among other factors, this is due to the accessibility of regional foods in rural areas. In contrast, urban areas show a marked consumption of ultra-processed foods due to the urbanization process. However, studies show that with the progress of urbanization and industrialization, in a few years the rural population will tend to imitate the same eating habits as urban areas³.

Deficits in knowledge, attitudes, and practices regarding regional foods directly limit their consumption. The current literature presents several sociodemographic and economic variables that can influence the population's KAP, such as age, race, religion, income, gender, and marital status⁶⁻⁷.

Despite advances in the field of food and nutrition, there are few studies that specifically investigate the factors that influence knowledge, attitude, and practice (KAP) regarding the use of regional foods, especially in families with children under five years of age. To date, the only study that has applied KAP to investigate regional foods in families with young children in Ceará is Silveira *et al.*⁸. Understanding these variables is essential to support food and nutrition education strategies and strengthen public policies aimed at promoting healthy habits in childhood.

The Knowledge, Attitudes, and Practices (KAP) survey stands out as an appropriate tool for this type of research, as it provides relevant data on a specific population in a given context. In addition, it allows for the assessment of areas (knowledge, attitudes, and practices) that require more effective interventions. Thus, the use of KAP is especially useful for identifying

ntialities related to the use of regional foods, providing evidence to guide more tions⁹.

In Brazil, despite public policies that encourage the use of regional foods as a strategy to promote food and nutritional security, there has been a progressive decline in the consumption of these foods, especially in urban areas. In the interior of the state of Ceará, in turn, there are still few studies investigating the sociodemographic factors that influence the use of regional foods in families with young children, making it difficult to formulate educational strategies and policies adapted to the local reality.

Therefore, it is necessary to investigate how sociodemographic and economic variables influence the KAP of regional food use. With this knowledge, it becomes possible to effectively promote the consumption of these foods, which are essential for the healthy growth of children. In addition to supporting public policies on child nutrition and food education initiatives.

This study aimed to evaluate the association of sociodemographic and economic factors with the levels of knowledge, attitude, and practice regarding the use of regional foods by families with children under five years of age in rural Ceará.

METHODOLOGY

This is a cross-sectional study that is part of a quasi-experimental research project entitled "Effect of the application of educational intervention to promote food and nutritional security in families with children," conducted in two Primary Health Care Units (PHCU) and two public preschools in the municipalities of Acarape/CE and Redenção/CE.

Data collection took place from January to March 2023, involving mothers of children under five years of age. The following inclusion criteria were adopted: being the mother of a child under five years of age, being registered and monitored at the UAPS where the study was conducted, and residing with the child in the same household. The exclusion criteria were: mothers who did not have the cognitive ability to participate in the educational intervention, as reported by the participant herself.

The population was selected through convenience sampling, considering accessibility to participants during the collection period. This approach was adopted due to the logistics of

eligible mothers registered at participating health units and schools, as well as and the team available for the study.

At the PHCU, data collection was carried out on childcare and vaccination days, when there were a greater number of mothers present. In preschools, data collection took place at the time when parents dropped off and picked up their children from school.

To collect data for the study, we first applied a questionnaire on the sociodemographic profile and health status of the child, which was adapted from the study by Joventino¹⁰, followed by the knowledge, attitude, and practice (KAP) survey used in a previous study by Martins², which made it possible to assess the adequacy or inadequacy of the use of regional foods in children's meals.

The Joventino¹⁰ form consists of 59 questions subdivided into sections on demographic data, socioeconomic and health conditions, and child health. The adaptations made consisted of removing questions related to health conditions and childhood diarrhea. At the end of the adaptations, the questionnaire was reduced to 21 closed questions that emphasize the sociodemographic data of the parents (participant's age, education, marital status, occupation, number of people in the household, family income) and the child's health.

As for the second instrument, the KAP survey was developed and used in the study by Martins². Previous studies addressing the survey topic were used in its development, ensuring its suitability for the regional foods topic¹¹⁻¹².

The KAP survey consists of three sections: (1) knowledge about regional foods, addressing recognition, purpose, and preparation methods; (2) attitudes toward the use of these foods in children's diets, investigating perceptions of need and justifications; and (3) eating habits, covering previous use and frequency of use of regional foods. The assessment of each domain was classified as adequate or inadequate, according to criteria established in the instrument itself.

To improve understanding of the second instrument, it is essential to outline the concepts of knowledge, attitude, and practice. Thus, knowledge can be characterized as the understanding of a specific phenomenon; attitude as beliefs and predispositions regarding a given scenario; and practice as the actions taken to achieve a specific goal².

ge was considered adequate when the informant reported having heard about new that they can be used to prepare a variety of foods, cited three types of foods, and mentioned at least two types of foods prepared with these foods, with the exception of juice. Knowledge was considered inadequate when the informant did not know what regional food was used for, or had heard of it but did not know that it could be used to prepare a variety of foods; when she did not mention vegetables, tubers, and fruits; or when she could not mention two types of food prepared with these foods².

With regard to attitude, this is considered adequate when the mother stated that it was necessary to use regional foods in the child's diet, with the justification of preventing malnutrition, obesity, among other nutritional disorders, and/or promoting food security and/or new food alternatives; inadequate when the informant did not consider it necessary or did not know how to answer about the use of regional foods or justify it in any other way².

The practice was classified as adequate when the informant had already used food containing regional components in the child's diet and could use this regional food whenever she deemed necessary. Inadequate practice occurred when the participant had never used food containing a regional ingredient in the child's diet and did not know that she could use this regional food whenever she deemed necessary, establishing limits on supply².

This study is part of a larger project entitled "Effect of the application of educational intervention to promote food and nutritional security for families with children," whose main focus was to assess the impact of educational actions on the knowledge, attitude, and practice of mothers with children under five years of age regarding the use of regional foods. In the context of this project, before the KAP survey was administered, participants were instructed on the concept of regional food through the serial album "Regional foods promoting food and nutritional security," which illustrates local examples and culinary practices based on these ingredients. This step aimed to ensure prior and standardized understanding among respondents regarding the key concept of the investigation.

The collected data were entered into Microsoft Excel 2010 and analyzed using the *Statistical Package for the Social Sciences (IBM SPSS Statistics)* program, version 20.0. The Kormogorov-Smirnov test with Lilliefors correction was performed, verifying that the data did not show a normal distribution ($p\text{-value} < 0.05$). To verify the association between

ic and economic variables and adequate knowledge, attitudes, and eating
ldren, logistic regression was performed using the “insert” method. The
magnitude of the effect of each variable was estimated by calculating the odds ratio and its
respective 95% confidence intervals and 5% significance level.

The study was approved by the Research Ethics Committee in accordance with opinion
No. 5,792,199 and CAAE No. 63788322.0.0000.5576. All standards for research involving
human subjects, as set forth in Resolution 466/2012 of the Brazilian National Health Council,
were complied with, and the participants signed an informed consent form to participate in the
study.

RESULTS

The sample consisted of 70 mothers of children under five years of age. Regarding KAP
domains, most participants had inadequate knowledge (82.9%), while 81.4% demonstrated an
adequate attitude and 75.7% reported adequate dietary practices regarding the use of regional
foods.

Table 1 shows the distribution of sociodemographic and economic variables in relation
to the participants' level of knowledge about the use of regional foods. Regarding education,
there was a statistically marginal association ($p = 0.057$), with participants with complete or
incomplete undergraduate degrees showing a higher frequency of adequate knowledge. Logistic
regression analysis indicated that these participants were 88.5% less likely to have inadequate
knowledge ($OR = 0.115$; 95% CI: 0.012–1.089) compared to those with elementary school
education.

Participants with partners had a higher percentage of inadequate knowledge (54.3%).
The odds ratio for this group was 5.789 (95% CI: 0.697–48.111), although not statistically
significant ($p = 0.092$). The occupation “housewife” stands out as the one with the highest
concentration of participants with inadequate knowledge (51.7%). For example, women with
occupations classified as “other” had a significant reduction in the odds of inadequate
knowledge ($OR = 0.104$; 95% CI: 0.023–0.467).

nts living with up to four people in the household had a higher percentage of dge (66.7%) compared to those living with more than four people (OR = 0.417; 95% CI: 0.105–1.656), although this was not statistically significant ($p = 0.240$).

The family income variable suggests that participants with an income higher than the minimum wage had a higher prevalence (58.3%) and a 67.9% greater chance of adequate knowledge than those with lower incomes (OR = 0.321; 95% CI: 0.090–1.151).

showed one of the highest percentages for adequate knowledge (17.1%), demonstrating a statistically significant association ($p\text{-value} < 0.05$) with the participants' knowledge. The analysis indicated that these mothers were 26.7% more likely to have adequate knowledge (OR = 0.733; 95% CI: 0.615–0.875), reinforcing the positive impact of health education initiatives.

iation of sociodemographic and economic variables and participants' level of
t the use of regional foods. Redenção, Ceará, Brazil, 2023

VARIABLES	Knowledge				Odds	p-value
	Adequate		Inadequate			
Participant age	N	%	N	%		
18 to 29 years	6	8,6	30	42,9	-	0,827
30 to 39 years old	5	7,1	20	28,6	0,625 (0,065 – 5,966)	
40 or more	1	1,4	8	11,4	0,500 (0,050 – 4,978)	
Education						
Complete and incomplete primary education	1	1,4	16	22,9	-	0,057
Complete and incomplete secondary education incomplete	5	7,1	31	44,3	0,388 (0,042 – 3,604)	
Complete and incomplete undergraduate degree	6	8,6	11	15,7	0,115 (0,012 – 1,089)	
Marital status						
With partner	11	15,7	38	54,3	5,789 (0,697 – 48,111)	0,092
No partner	1	1,4	20	28,6		
Occupation						
Homemaker	3	4,3	36	51,4	1	0,035*
Student	1	1,4	4	5,7	0,333 (0,28 – 4,011)	
Day laborer (cleaner)	0	0,0	2	2,9	-	
Seamstress	0	0,0	1	1,4	-	
Self-employed	0	0,0	3	4,3	-	
Salesperson	0	0,0	2	2,9	-	
Others	8	11,4	10	14,3	0,104 (0,023 – 0,467)	
Number of people in the household						
Less than 4 people	8	66,7	48	82,8	0,417 (0,105 – 1,656)	0,240
More than 4 people	4	33,3	10	17,2		
Family income "						
Less than minimum wage	5	41,7	40	69,0	0,321 (0,090 – 1,151)	0,100
Higher than minimum wage	7	58,3	18	31,0		
Does the child have any illness?						

Source: Survey data, 2023.

g attitudes toward the use of regional foods, the occupation "homemaker" ically significant association with the appropriate attitude ($p = 0.035$), and a high prevalence was also observed among participants who had a partner, who were 2.4 times more likely to have the appropriate attitude ($OR = 2.4$; 95% CI: 0.695–0.829), although this was not statistically significant ($p = 0.188$). The highest prevalence of positive attitudes was also noted among women with lower family income, although this was not statistically significant (Table 2).

ciação das variáveis sociodemográficas e econômicas e o nível de atitude dos re o uso dos alimentos regionais. Redenção, Ceará, Brasil, 2023

VARIABLES	Attitude				Odds	P-value
	Adequate N	%	Inadequat N	%		
Participant age						
18 to 29 years	26	37,1	10	14,3	1	0,123
30 to 39 years old	23	32,9	2	2,9	0,226 (0,045 – 1,141)	
40 or more	8	11,4	1	1,4	0,325 (0,036 – 2,942)	
Education						
Complete and incomplete primary education	12	17,1	5	7,1	1	0,063
Complete and incomplete secondary education incomplete	28	40,0	8	11,4	0,686 (0,186 – 2,531)	
Complete and incomplete undergraduate degree	17	24,3	0	0,0	-	
Marital status						
With partner	42	60,0	7	10,0	2,4 (0,695 – 8,290)	0,188
No partner	15	21,4	6	8,6		
Occupation						
Homemaker	30	42,9	9	12,9	1	0,035*
Student	5	7,1	0	0,0	-	
Day laborer (cleaner)	2	2,9	0	0,0	-	
Seamstress	0	0,0	1	1,4	-	
Self-employed	1	1,4	2	2,9	6,667 (0,540 – 82,310)	
Salesperson	2	2,9	0	0,0	-	
Others	17	24,3	1	1,4	0,196 (0,23 – 1,683)	
Number of people in the household						
Less than 4 people	45	64,3	11	15,7	0,682 (0,133 – 3,500)	1,000
More than 4 people	12	17,1	2	2,9		
Family income "						
Less than minimum wage	36	51,4	9	12,9	0,762 (0,209 – 2,781)	0,759
Higher than minimum wage	21	30,0	4	5,7		
Does the child have any illnesses?						

Source: Survey data, 2023.

Proper eating habits were observed more frequently among younger participants and those with a high school education. Having a partner was an important factor, with participants

ner being 2.7 times more likely and having a higher percentage of proper 2.7; 95% CI: 0.875–8.548), although this was not statistically significant ($p = 0.126$) in relation to the use of regional foods (Table 3).

There was a statistically significant association with the occupation “Homemaker” ($p < 0.05$), which was the group with the highest prevalence of positive practices. In addition to this factor, having a partner and living with up to four people in the household showed a tendency to be associated with adequate practices, although without statistical significance.

Participants whose children did not have any diseases had a higher percentage of

information about the importance of proper nutrition for children was associated with a 2.6 times greater likelihood of proper practices (OR = 2.6; 95% CI: 0.850–7.961) in relation to the use of regional foods when compared to participants who did not receive this information (Table 3).

iation of sociodemographic and economic variables and the level of practice of
rding the use of regional foods. Redenção, Ceará, Brazil, 2023

VARIABLES	Practice				Odds	P-value
	Adequate N	%	Inadequate N	%		
Participant age						
18 to 29 years	28	40,0	8	11,4	1	0,419 ¹
30 to 39 years old	17	24,3	8	11,4	1,647 (0,521 – 5,204)	
40 or more	8	11,4	1	1,4	0,438 (0,047 – 4,038)	
Education						
Complete and incomplete primary education	12	17,1	5	7,1	1	0,126 ¹
Complete and incomplete secondary education incomplete	25	35,7	11	15,7	1,056 (0,299 – 3,728)	
Complete and incomplete undergraduate degree	16	22,9	1	1,4	0,150 (0,015 – 1,457)	
Marital status						
With partner	40	57,1	9	12,9	2,735 (0,875 – 8,548)	0,126 ²
No partner	13	18,6	8	11,4		
Occupation						
Homemaker	30	42,9	9	12,9	1	0,008 ^{1*}
Student	5	7,1	0	0,0	-	
Day laborer (cleaner)	1	1,4	1	1,4	3,333 (0,189 – 58,799)	
Seamstress	0	0,0	1	1,4	-	
Self-employed	0	0,0	3	4,3	-	
Salesperson	1	1,4	1	1,4	3,333 (0,189 – 58,799)	
Others	16	22,9	2	2,9	0,417 (0,080 – 2,165)	
Number of people in the household						
Less than 4 people	40	57,1	16	22,9	0,192 (0,023 – 1,594)	0,162
More than 4 people	13	18,6	1	1,4		
Family income "						
Less than minimum wage	34	48,6	11	15,7	0,976 (0,311 – 3,059)	1,000 ²
Higher than minimum wage	19	27,1	6	8,6		
Does the child have any illnesses?						

*p value < 0.05.

Source: Data from the study itself, 2023

shows a strong association and statistical significance between the domains of knowledge, attitude, and practice. No mother with adequate knowledge presented inadequate practice; and those with the appropriate attitude were 5.4 times more likely to have adequate practice (OR = 5.48; 95% CI: 1.515–19.849), reinforcing the importance of positive attitudes in children's eating behavior.

Table 4 – Association between the domains of knowledge and attitude with the domain of practice. Redenção, Ceará, Brazil, 2023.

Domains	Practice				Odds	p-value
	Adequate N	%	Inadequate N	%		
Knowledge						
Adequate	12	17,1	0	0,0	1,415 (1,199 – 1,670)	0,031**
Inadequate	41	58,6	17	24,3		
Attitude						
Adequate	47	67,1	10	14,3	5,483 (1,515 – 19,849)	0,011**
Inadequate	6	8,6	7	10,0		

¹ Fisher's exact test, *p-value < 0.05.

Source: Data from own research, 2023

DISCUSSION

The results of this study suggest that variables such as occupation and receipt of information about proper nutrition have an important influence on caregivers' eating habits. These results also indicate that interventions to improve caregivers' knowledge and attitudes may be effective in promoting proper eating practices among caregivers in relation to feeding children regional ingredients.

When it comes to the use of regional foods, nutritional knowledge is an important predictor, given that understanding of the subject favors appropriate behaviors regarding the choice of these foods¹³.

Given the results of this study, participants with a complete primary education were less likely to have inadequate knowledge about regional foods, indicating that knowledge is not restricted to levels of education. e, scientific, technical, and popular knowledge, thus combating the hegemonic knowledge that deprives scientific construction of its diversity¹⁴.

ating the analyses of other studies on nutritional literacy, individuals with
ave greater nutritional knowledge¹³. Regarding the receipt of information on
healthy eating for children, participants were more likely to have adequate knowledge than
were 2.5 times more likely to have an appropriate attitude and 2.6 times more likely to have
appropriate practices. This result reinforces the importance of health awareness and education
campaigns as tools for improving knowledge, attitudes, and practices. appropriate infant
foods¹⁵.

In addition, marital status "with partner," more than four people living in the same
household, and mothers of children "without diseases" were screened because they were
associated with inadequate knowledge parameters.

It should be noted that maternal occupation as a homemaker was significantly associated
with inadequate knowledge ($p < 0.05$). In the 20th century, there were a considerable number
of women who were engaged in domestic work and had low levels of education, ranging from
those who could only sign their names to those who could not read or write. Despite the
encouragement of literacy among domestic workers, resulting from social struggles, this social
class still has the lowest levels of education, a factor that influences levels of knowledge, even
on everyday topics such as regional foods¹⁶.

At first glance, it was noticeable that mothers with higher levels of education had more
appropriate attitudes. Attitude is understood as beliefs and opinions about a particular object or
situation, so levels of education are directly associated with individuals' understanding of the
importance of a healthy diet and the inclusion of regional foods in their menu¹⁷.

Appropriate practice in the use of regional foods refers to the inclusion of these
components in children's diets. National studies point to a steady decline in the consumption of
regional ingredients by Brazilians. This reality is reflected in risky eating patterns, given that
typical foods, whether fresh or minimally processed, are processed foods are being replaced by
processed and ultra-processed foods, and this condition also contributes to the loss of culinary
identity¹⁸.

Among the sociodemographic variables analyzed, occupation showed a statistically
significant association ($p\text{-value} < 0.05$) with adequate practices. The occupation " homemaker"

st percentages (42.9%) of adequate practice in the use of regional foods. This s higher levels of food security and is associated with greater availability of time and dedication of these mothers to preparing quality meals with seasonal ingredients. Research indicates that individuals in occupations with long hours outside the home tend to rely more on processed foods or fast food, which can lead to a decrease in diet quality; while individuals who prepare meals at home, such as "housewives," generally have a greater ability to prepare meals with fresh and less processed ingredients, promoting healthier and more consistent eating habits¹⁹.

According to the results of this study, participants with a household income below the minimum wage reported adequate practices, which may be associated with the fact that regional foods are economically favorable and easily accessible. This characteristic makes them a viable alternative for families in situations of socioeconomic vulnerability, which explains the increase in consumption of these foods by this population.

On the other hand, low family income can also be a significant barrier to the adoption of healthy eating practices. Research shows that low-income individuals have limited time to prepare meals and gaps in their knowledge of how to do so, which directly affects the quality of their diet²⁰.

As in other studies, participants with higher levels of education showed a preference for healthy eating habits, and in the context of this study, they opted for tubers, legumes, and regional fruits. Despite these results, higher education does not guarantee a healthy diet, given that there is a considerable percentage of individuals with the same educational profile who base their food choices on fast food. This divergence in dietary patterns in the same individual is called a "dual dietary pattern" and shows that knowledge does not always translate into practice²¹.

Finally, it should be noted that the data from this study prove that adequate levels of knowledge and attitude directly reflect adequate practices, increasing the consumption of regional foods and generating healthy eating patterns.

The Food Acquisition Program (FAP) aims to encourage family farming and the consumption and appreciation of food produced by this type of agriculture. In this way, it ensures food in the necessary quantity, quality, and regularity, guaranteeing the human right to

althy food²². Considering the benefits of consuming regional foods and the , it is essential to spread knowledge and use of these foods, taking into account sociodemographic and economic factors.

The study has the limitation that sampling was performed for convenience, although operationally feasible, which restricts the representativeness of the general population and, consequently, the generalization of the results. This approach was chosen due to the ease of access to the target audience of this study. In addition, the small sample size may not capture all the variability of the sociodemographic factors present in the study population, which may affect statistical power.

The possibility of response bias should also be noted, since the data were self-reported by the participants, which may influence the veracity or accuracy of the information provided. Thus, the findings should be interpreted with caution when extrapolating the results, and indicate the importance of future studies with larger probabilistic samples in different territorial and cultural contexts. The findings of this study reinforce the need to consider sociodemographic factors, such as occupation, income, and access to information, in the formulation of public policies aimed at child feeding and nutrition. The identification of greater

nutritional guidance points to the potential of educational actions in Primary Health Care.

Programs such as the National School Feeding Program (NSFP) and the Food Acquisition Program (FAP) can be strengthened by incorporating regional strategies that promote the consumption of typical local foods, whose concrete interventions include the training community health workers and education professionals to develop food education activities focused on regional foods, as well as the inclusion of practical workshops for preparing these foods in health centers and schools, and coordination with local producers to facilitate access.

Such strategies, if well implemented, can contribute to the food and nutritional security of vulnerable children, valuing local food culture and reducing exposure to the consumption of ultra-processed foods.

ge, attitudes, and practices regarding regional foods among families with children under five years were significantly associated with maternal occupation and receipt of nutritional guidance; knowledge is also influenced by the receipt of health information on child nutrition. The results suggest that sociodemographic and economic factors influence the CAP of the use of these foods.

Given the results obtained, the importance of educational interventions is reinforced, considering the sociodemographic context, on the promotion of healthy eating habits in early childhood through the use of regional foods. Although this study did not directly measure the consumption of ultra-processed foods, the results point to the importance of promoting regional foods as a strategy to mitigate this consumption and thus reduce the risks of chronic noncommunicable diseases, such as anemia, overweight, and underweight in childhood.

Given the scarcity of literature, it is recommended that new longitudinal and comparative studies be conducted in different socioeconomic contexts, with representative samples, to assess the association of sociodemographic factors and levels of knowledge, attitude, and practice regarding the use of regional foods in different cultural and territorial contexts. These studies could identify predictive factors and guide culturally sensitive food promotion strategies.

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